Seestar Toolkit Quick Start

Development — ENH-006 (unreleased)

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This development edition includes the unreleased ENH-001 configuration commands, ENH-002 observing-night date policy, ENH-003 archive light-TIFF option, and ENH-005 FITS privacy commands, and ENH-006 archive dry-run reporting. They are not included in the published v1.1.0 wheel. The released installation instructions and support boundaries remain as described below. No new release number has been assigned.

This Quick Start gets you from installation to a **safe first archive** and, optionally, a FITS-to-TIFF conversion.

It assumes you are comfortable using Terminal and Homebrew. For explanations, advanced archive options, troubleshooting and the full v1.1.0 support boundaries, see the Seestar Toolkit User Guide.

For your first archive:

keep a backup
use `--dry-run`
inspect the plan
copy first
verify the result
only consider `move` later

1. Check the requirements

Seestar Toolkit v1.1.0 is officially supported on:

- **Seestar S50** captures validated for this release;
- an **Apple-silicon Mac** using a supported macOS version;
- **Python 3.11, 3.12, 3.13 or 3.14**.

Other telescope models, operating systems, computer architectures and Python versions may work, but are not supported unless specifically listed for this release.

Solar, Lunar and Planetary video processing and DSLR capture processing are outside the scope of v1.1.0.

Check Homebrew:

```
/opt/homebrew/bin/brew --version
```

See which Homebrew Python versions are installed:

```
/opt/homebrew/bin/brew list --formula | grep '^python'
```

For example, check Python 3.13:

```
/opt/homebrew/bin/python3.13 --version
```

If you need to install it:

```
/opt/homebrew/bin/brew install python@3.13
```

Do not modify Apple's system Python.

2. Create the Toolkit environment

Create the recommended virtual environment:

```
/opt/homebrew/bin/python3.13 -m venv ~/.venvs/seestar-toolkit
```

Activate it:

```
source ~/.venvs/seestar-toolkit/bin/activate
```

Check Python:

```
python --version
```

You create this environment once. In later Terminal sessions, simply activate it again; you do **not** need to reinstall the Toolkit each time.

3. Install Seestar Toolkit

These steps install the **published v1.1.0 release**, not ENH-001. The new configuration commands require an installation containing the development changes; confirm `seestar-toolkit config --help` is available in the environment you intend to use. The package version may still report `1.1.0`, so that number alone does not identify an ENH-001 build.

Download the v1.1.0 release package from the project's GitHub Releases page.

The recommended installation file is:

```
seestar_toolkit-1.1.0-py3-none-any.whl
```

Seestar Toolkit v1.1.0 is **not distributed through PyPI**.

With the virtual environment active, install the wheel, replacing the example path with the real downloaded file:

```
python -m pip install \
  "/path/to/downloads/seestar_toolkit-1.1.0-py3-none-any.whl"
```

Check the installation:

```
python -m pip check
seestar-toolkit --version
```

The version command should report:

```
seestar-toolkit 1.1.0
```

You can also check the equivalent Python entry point:

```
python -m seestar_toolkit --version
```

Normal use in this guide uses the `seestar-toolkit` command.

4. Choose a safe first archive

Start with a **small Seestar observing session that is already backed up**.

You need:

- a **source root** containing the Seestar captures;

- a separate **archive root** where the organised archive will be created.

Archive discovery is shallow: the Toolkit examines the selected source root and its immediate child directories. It does not recursively search an arbitrary folder tree.

A typical Seestar work folder can look like:

```
My Works/                          <- select this as SOURCE_ROOT
  IC 434/
    Stacked_195_IC 434_10.0s_LP_20260103-225603.fit
  IC 434_sub/
    Light_IC 434_10.0s_LP_20260103-220259_thn.jpg
    Light_IC 434_10.0s_LP_20260103-220259.fit
    Light_IC 434_10.0s_LP_20260103-220259.jpg
```

Select `My Works` itself. JPEGs and thumbnails alongside the FITS captures are normal; the archive workflow leaves them untouched.

Keep the archive root outside the source tree.

5. Preview before writing anything

For the first run, explicitly use **copy** and **dry-run**.

Replace `Home` and both example paths:

```
seestar-toolkit archive --dry-run --source-action copy \
  --collision-policy skip-identical --location "Home" \
  --non-interactive "/path/to/My Works" "/path/to/archive"
```

`--dry-run` previews the proposed archive without writing archive outputs. Its normal report gives one compact block per planned observation, including the resolved destination, telescope metadata and planned light/stack FITS counts. Use global `--verbose` before `archive` when you also need the detailed source-to-destination FITS mappings:

```
seestar-toolkit --verbose archive --dry-run SOURCE_ROOT ARCHIVE_ROOT
```

Before continuing, check that the preview has the expected:

- captures;
- target;
- location;
- observing night;
- observation grouping;
- archive destination;
- `copy` source action.

If anything looks wrong, stop and investigate while you are still in dry-run.

6. Create the archive by copying

When the preview looks correct, run the same command with only `--dry-run` removed:

```
seestar-toolkit archive --source-action copy \
  --collision-policy skip-identical --location "Home" \
  --non-interactive "/path/to/My Works" "/path/to/archive"
```

The default archive hierarchy keeps a stack TIFF companion beside its stack FITS:

```
Target/
└── Location/
    └── Observing night/
        └── observation_01/
```

```
├── fits_images/
│   └── lights/
└── seestar_stacked/
```

The Toolkit may create additional observation directories when the captures represent separate observations. A human-readable target index is also maintained.

Because this example uses `copy`, the source FITS files remain in their original location.

7. Verify the archive

Before doing anything to your source collection, inspect both the source and the new archive.

Confirm that:

1. the original source captures are still present;
2. the expected target, location and observing-night directories exist;
3. the expected observation directories and files are present;
4. the stack TIFF companion and target index are where expected;
5. the command did not report unresolved observations, collisions or failures.

Do **not** delete your originals simply because the command completed.

The full User Guide explains observing-night dates, observation reconstruction, custom hierarchy order, collisions, re-running an archive and the additional risks of `--source-action move`.

8. Save preferences and observing locations

ENH-001 development build required. The published v1.1.0 release only reads manually created configuration and does not provide these `config` commands.

Saved preferences and locations use:

```
~/.config/seestar-toolkit/config.toml
```

Inspect the values that would normally apply:

```
seestar-toolkit config show
```

Each preference appears once, labelled **saved** or **default**. Saved locations are included. `config show --saved` shows saved entries only; `config show --defaults` shows built-in defaults without reading the saved file. Inspection does not create or change configuration.

To save a preference explicitly:

```
seestar-toolkit config set archive.source_action copy
```

The first valid save can create the absent default file. Command-line overrides still win for an individual archive run. To remove this saved preference and return to the built-in default:

```
seestar-toolkit config unset archive.source_action
```

Inspect a capture, then save its site

First read a suitable capture's site coordinates and compare them with any saved locations:

```
seestar-toolkit config show --extract "/path/to/capture.fit"
```

This saves nothing. It shows every matching site's name, coordinates, radius and distance, or reports that no saved site matches.

To save those coordinates as `Home`:

```
seestar-toolkit config set --location "Home" --extract "/path/to/capture.fit"
```

A **new** location defaults to a **100 m radius**. Add `--radius-m 150`, for example, to choose another radius. All new or edited matching areas must be distinct: touching or overlapping circles are rejected.

If `Home` already exists, the command proposes an update and retains its saved radius unless you explicitly supply another. It shows the old/proposed values and asks **Update config / Skip**; pressing Enter without an answer skips. For an intended unattended replacement, add `--update`. This requires an existing entry and does not bypass validation.

These examples are alternatives, not commands to run as a single script. Replace the paths and location name with your own. See the User Guide's configuration chapter for manual coordinates, partial edits, rename, removal and repair rules.

Check automatic archive matching

An explicit `archive --location "Home"` overrides automatic matching. To check your saved site's matching, omit that override during a dry run:

```
seestar-toolkit archive --dry-run --source-action copy \
  --collision-policy skip-identical --non-interactive \
  "/path/to/My Works" "/absolute/path/to/archive"
```

Inspect the proposed paths. `M 42/Home/20260915/...` uses the saved site; `M 42/unknown/20260915/...` means the location was not resolved. Do not proceed with copying until the plan is correct. Matching is local, with no online lookup.

Archive-prompt answers are temporary; use explicit `config set` to save a site. Renaming/removing a saved site does not rename or remove an existing archive.

If a saved hierarchy is invalid, you can replace it with `config set` or remove it with `config unset`. See the User Guide for the repair steps and the separate handling of a broken configuration-file format.

9. Convert a FITS image to TIFF

Conversion is optional. You do not need to convert FITS files before using the archive.

Convert one supported FITS image:

```
seestar-toolkit convert \
  "/path/to/input.fit" \
  "/path/to/output.tiff"
```

For a light directly under a current archive's `observation_NN/fits_images/lights/`, omit the output to create the same-stem TIFF under `observation_NN/tiff_images/lights/`:

```
seestar-toolkit convert \
  "/path/to/archive/Target/Home/20260915/observation_01/fits_images/lights/Light_M42.fit"
```

Batch-convert the supported FITS files directly inside one directory:

```
seestar-toolkit convert-batch \
  "/path/to/fits" \
  "/path/to/tiff"
```

Batch conversion is non-recursive.

For a current archive observation, the batch input may be the observation, its `fits_images` directory or its `fits_images/lights` directory. All three scan only the canonical lights directory. Omit the output to infer `observation_NN/tiff_images/lights/`:

```
seestar-toolkit convert-batch \
  "/path/to/archive/Target/Home/20260915/observation_01"
```

Supplying an explicit batch output still resolves any of those three archive input levels to canonical lights. Outside a recognised current archive, both commands require their explicit output form. A missing canonical lights directory is an error; an empty one reports no eligible FIT/FITS files and a failed batch.

The Toolkit creates **linear RGB TIFF** data. It does not automatically stretch the image or alter brightness, contrast or colour, so an astronomical TIFF can initially look very dark.

An inferred `tiff_images/lights/` directory is created only when a valid conversion needs it. Empty batches and invalid inputs do not create it. Existing TIFF destinations are never overwritten; batch conversion continues with independent inputs and reports `SUCCESS`, `PARTIAL SUCCESS` or `FAILED`. Complete success returns `0`; partial or failed processing and inference errors return `1`; malformed command usage returns `2`.

Keep the original FITS file.

10. Useful commands

Show the main help:

```
seestar-toolkit --help
```

Show command-specific help:

```
seestar-toolkit archive --help
seestar-toolkit convert --help
seestar-toolkit convert-batch --help
seestar-toolkit header-report --help
seestar-toolkit pii-scrub --help
```

In an ENH-001 installation, also use:

```
seestar-toolkit config --help
seestar-toolkit config show --help
seestar-toolkit config set --help
seestar-toolkit config unset --help
```

Check the installed version:

```
seestar-toolkit --version
```

The principal exit codes are:

```
0    success
1    operational or processing failure
2    command-line usage error
```

A batch or archive failure does not mean that every earlier successful action has automatically been rolled back. Inspect the source and destination before retrying.

11. When you have finished

Leave the virtual environment:

```
deactivate
```

Your Toolkit installation remains in:

```
~/.venvs/seestar-toolkit/
```

Activate it again next time:

```
source ~/.venvs/seestar-toolkit/bin/activate
```

Next: the full User Guide

This development Quick Start deliberately leaves out many details. Final ENH-001 validation remains pending.

Read the Seestar Toolkit User Guide before using `move`, changing collision policies, customising the archive hierarchy, troubleshooting partial operations, or working through storage, permissions and privacy questions.

In particular, remember:

Back up → Dry-run → Inspect → Copy → Verify

Your original FITS captures remain valuable source data. Keep an independent backup and verify your archive before deciding whether to remove source files.

Before sharing FITS files, configuration, archive indexes or diagnostics, review them for private location, device and capture metadata.